UNIFORM ASYMPTOTICS FOR A MULTIDIMENSIONAL RENEWAL RISK MODEL WITH RANDOM NUMBER OF DELAYED CLAIMS AND MULTIVARIATE SUBEXPONENTIALITY

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Abstract. In this paper we examine a multivariate risk model, with common renewal counting process, constant interest rate, and each claim vector is accompanied by a random number of delayed claim vectors. The interest is focused on the asymptotic behavior of the entrance probability of the discounted aggregate claims into some rare-sets, over a finite and an infinite time horizon. Our results study the case where the main claims and the delayed claims have in some sense, asymptotic equivalent tails, but also the case where the delayed claims are negligible with comparisons with the main claims. More precisely, our estimations over finite time horizon are equipped with local uniformity, and are valid under the assumption of class $\mathcal{S}_A$, of multivariate subexponential distributions for the claim distributions. On the case of infinite time horizon we need a mild restriction on the distribution class $\mathcal{A}_A^*$ of multivariate subexponential distributions with positive lower Karamata index. The asymptotic relations reflect completely as all the sources of randomness, under the concrete rare-sets A, and the different dependence structures as well, without loosing elegance in spite of their generality. Further, we provide some more explicit formulas, together with relaxations of some assumptions, for the claim distributions from the multivariate regular variation. For the proof of the main results on infinite time case and for the construction of examples of multivariate distributions we need some closure properties of distribution classes $\mathcal{A}^*$ and $\mathcal{A}_A^*$. Especially, we present some necessary and sufficient conditions for the closure property with respect to convolution and some sufficient conditions for the closure property with respect to product convolution. Finally, we carry out some numerical studies to show the accuracy of our asymptotic estimations. The examples contain also cases with moderate heavy tails.

Keywords: Discounted aggregate claims; Uniformity; Infinite time horizon; Lower Karamata index; Closure properties; Multivariate linear single big jump principle

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